Supplementary Material: Exact Radial-Determinant Frequencies of Annular and Circular Plates: the Closed-Ring and Solid-Disk Limits of the Sector Wave-Function Method

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This file is the complete Supplementary Material to the companion manuscript, §§S.1–S.6. The main text's §2.1 cites §S.1; §2.2 and §5.1 cite §S.2; §2.4 cites §S.3; §§4.1, 6.1 and 7 cite §S.4; job-level provenance for every reported sweep is §S.5; and §6.3 cites §S.6. The two files compile separately, so cross-references here are by section rather than by table number.

S.1 Bessel span of the four Frobenius branches

The radial content of the sector assembler is a four-term Frobenius series generated from mid-radius Taylor initial conditions rather than from named special functions. That choice is deliberate — it is what lets the same code carry a sector, a ring and (after the projection of §S.2) a disk — but it means the connection to the classical Bessel description has to be stated rather than assumed.

For a uniform isotropic Kirchhoff plate in polar coordinates, separation with $w(r, \theta, t) = W(r) \cos n\theta \, e^{i\omega t}$ reduces the plate equation to

$$\left(\nabla_n^2 + k^2\right)\left(\nabla_n^2 - k^2\right) W = 0, \qquad \nabla_n^2 = \frac{\mathrm{d}^2}{\mathrm{d}r^2} + \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r} - \frac{n^2}{r^2}, \qquad k^4 = \frac{\rho h \omega^2}{D}, \tag{S.1}$$

whose general solution is the four-dimensional space

$$W(r) = A\, J_n(kr) + B\, Y_n(kr) + C\, I_n(kr) + E\, K_n(kr). \tag{S.2}$$

The four columns returned by the series routine at a given (n, Ω) are a basis of that same four-dimensional space, expressed in the stretched radial variable x used throughout the series, not in $\{J_n, Y_n, I_n, K_n\}$ individually. Two consequences are used in the main text.

First, on a closed ring both arcs are genuine boundaries, so all four coefficients survive, the characteristic matrix is 4×4, and its rows are the two edge quantities at each arc as listed in the main text. The propagating pair $\{J_n, Y_n\}$ and the evanescent pair $\{I_n, K_n\}$ are both required; truncating to either pair is not an approximation of the ring problem but a different problem.

Second, Y_n and K_n are singular at $r = 0$ while J_n and I_n are regular there and behave as $r^{|n|}$. On a solid disk the two singular branches are inadmissible, which is why two coefficients survive, and it is that $r^{|n|}$ behaviour — not the numerical size of W near the origin — that identifies the surviving pair. §S.2 makes that identification constructive.

S.2 Disk regularity, the inner jet, and the rank-3 trap

S.2.1 Why "smallest $|W|$ at the origin" fails

The obvious way to isolate the regular pair is to evaluate the four columns near $r = 0$ and discard the two with the largest magnitude. This does not work, and the reason is worth recording because the failure is silent. The series columns are *truncated*: every one of them is finite at $r = 0$, including the two whose exact continuations are Y_n and K_n. A singular-value decomposition of the four columns evaluated at the origin therefore returns a mixture, and the 2×2 built on that mixture has no zero in the neighbourhood of the published frequency. This was diagnosed by observing that the resulting 2×2 determinant had no sign change at all across a window in which the published disk root is known to lie.

S.2.2 The inner jet

The correct discriminator is the local behaviour, not the local magnitude. Regular solutions of (S.1) behave as $r^{|n|}$ near the origin, which supplies two independent homogeneous conditions on the jet of W at $r = 0$:

$$n = 0: \quad W'(0) = W'''(0) = 0; \qquad n = 1: \quad W(0) = W''(0) = 0; \qquad n \geq 2: \quad W(0) = W'(0) = 0. \tag{S.3}$$

Each line is two linear functionals on the four-dimensional solution space, so each defines a 2×4 matrix whose null space is two-dimensional. That null space *is* the regular pair, expressed in the series basis. Writing $C \in \mathbb{C}^{4\times 2}$ for a basis of it, the production disk operator is the projection of the two *outer*-edge rows of the ring 4×4 onto that pair,

$$L_{ij}^{\text{disk}} = \sum_{q=1}^{4} L_{r_i q}^{\text{ring}} C_{qj}, \qquad r_1 = 1 \ (M_r \text{ outer}), \quad r_2 = 3 \ (V_r \text{ outer}), \tag{S.4}$$

and disk frequencies are the zeros of $\det L^{\text{disk}}$. The parity of (S.3) is the parity of $r^{|n|}$: even $n = 0$ kills the odd derivatives, odd $n = 1$ kills the even ones, and from $n = 2$ upward both the value and the slope vanish.

The inner rows of the ring matrix — M_r and V_r at the inner arc — must *not* be used in place of (S.3). They are boundary conditions for a hole; at $R_i = 0$ there is no hole and they are not regularity conditions. Using them is the trap of §S.2.3.

S.2.3 The rank-3 trap

Evaluating the ring 4×4 at $R_i = 0$ and searching its determinant is the natural shortcut and it produces numbers that look plausible. The main text tabulates the ordered $\log_{10}$ singular values of that matrix; the pattern is that at $n = 0$ two singular values collapse (rank 2, a two-dimensional kernel, which is what a correct disk operator would have) but at $n = 1$, 2 and 3 only one collapses (rank 3, a one-dimensional kernel). A one-dimensional kernel cannot represent the two-parameter regular family, and the determinant zeros of that matrix are not the plate's frequencies. The production code raises an exception rather than returning a value on this path, and a regression test asserts that the disk entry point produces a 2×2 and never a 4×4.

S.2.4 The tiny-hole control

An independent check on the projection (S.4) is to run the *ring* 4×4 at a very small hole, $b/a = 10^{-3}$, and compare with the 2×2 at a pre-registered 10^{-3} relative tolerance. Two of the four rows fail that bar, and the control is reported in the main text as a named FAIL rather than being tuned into a pass. The interpretation is supported by the ladder: both failing rows classify NOT while both passing rows classify CLEAN, and the hole roots are genuine deep minima. A hole of one part in a thousand is a physically different plate from a solid one at the 10^{-3} level for the two lowest branches; it is not evidence that the 2×2 has missed a frequency, which the five-figure hold-out and the finite elements independently rule out.

S.3 Conversion algebra

Two conversion constants are used in this paper and neither is universal.

S.3.1 Flexural

The literature parameter is $\lambda^2 = \omega a^2 \sqrt{\rho/D}$ with $a = R_o$, $D = Eh^3/12(1-\nu^2)$ and h the full thickness. The solver's native Ω is a different nondimensionalisation, and the map between them is recomputed from the geometry and material of each run rather than stored. Concretely, the map is linear, $\lambda^2 = \kappa\,\Omega$, with κ obtained by evaluating the conversion at $\Omega = 1$.

The worked case is the free–free steel anchor plate of [1]: $R_o = 8$ m, full thickness 0.08 m, $b/a = 0.5$. There $\kappa = 4\pi^2 = 39.4784\ldots$, and the anchor's mode 7 at native $\Omega = 1.369611$ maps to $\lambda^2 = 54.0755$, which agrees with the finite-element value for that mode to 0.01%. This is a unit test in the regression suite, not a derivation.

The number $4\pi^2$ is an accident of that particular geometry and material and is not an identity. At $b/a = 0.3$ with the same material the factor is a different number; reusing 39.478 there produces a consistent-looking table that is wrong by several percent on every row. The rule enforced in code is that native Ω is never printed under a λ^2 label and that κ is never cached across geometries.

S.3.2 In-plane

Irie, Yamada and Muramoto define their in-plane parameter from the *extensional* rigidity $D_{\text{ext}} = Eh/(1-\nu^2)$ as $\lambda^2 = \rho h a^2 \omega^2 / D_{\text{ext}}$, so

$$\lambda = \omega a \sqrt{\rho(1-\nu^2)/E}. \tag{S.5}$$

This is dimensionally and numerically unrelated to the flexural λ^2: it is linear rather than quadratic in ωa, and it carries E and $(1-\nu^2)$ rather than the bending rigidity. Applying the flexural factor to an in-plane root, or vice versa, is a silent error that produces plausible magnitudes, so the two conversions are separate functions in the code with no shared cache.

The worked case is $\beta = 0.2$, $n = 0$ radial, first mode: native $\Omega = 0.775412$ maps through (S.5) to $\lambda = 1.80147$ against Irie's printed 1.802, a relative miss of 3.0×10^{-4}.

S.4 Full match lists

The main text reports each table at the Poisson ratio appropriate to its caption. This section carries the second Poisson ratio for the two unlabelled SP-160 tables, and the full in-plane list.

S.4.1 SP-160 Table 2.35 at $\nu = 1/3$

Twenty of the forty cells are MATCH and twenty are WEAK; none is a MISS, so this half contributes 40/40 to the 80/80 reported in the main text. The largest miss is 2.220% at $(2,1)$, $b/a = 0.9$. The WEAK cells sit on the $(2,0)$ and $(3,0)$ branches, whose printed values coincide with SP-160 Table 2.34 — Raju's explicitly $\nu = 1/3$ set — rather than with Vogel's Table 2.35. Table 2.35 itself prints no Poisson ratio, which is why both were run and why the paper-facing comparison is the $\nu = 0.30$ column.

Table S.1: Free–free annulus, $\nu = 1/3$, against SP-160 Table 2.35. 20 MATCH, 20 WEAK, no MISS.

	$b/a = 0.1$		$b/a = 0.3$		$b/a = 0.5$		$b/a = 0.7$		$b/a = 0.9$	
(n, s)	printed	solver	printed	solver	printed	solver	printed	solver	printed	solver
$(2, 0)$	5.30	5.1996	4.91	4.8205	4.28	4.2071	3.57	3.5234	2.94	2.8930
$(3, 0)$	12.4	12.220	12.26	12.060	11.4	11.261	9.86	9.7336	8.14	8.0429
$(0, 1)$	8.77	8.8188	8.36	8.3161	9.32	9.2288	13.2	13.018	34.9	34.429
$(1, 1)$	20.5	20.444	18.3	18.170	17.2	16.943	22.0	21.519	55.7	54.657
$(2, 1)$	34.9	34.929	33.0	32.907	31.1	30.698	37.8	37.083	93.8	91.718
$(3, 1)$	53.0	52.871	51.0	50.944	47.4	46.982	55.7	54.607	135.0	132.30
$(0, 2)$	38.2	38.169	50.4	50.222	92.3	92.209	251.0	250.51	2238.0	2238.8
$(1, 2)$	59.0	59.063	58.8	58.456	96.3	96.044	253.0	252.97	2240.0	2240.7

S.4.2 Mixed edges at $\nu = 1/3$

Table S.2 is the inner-free / outer-clamped set and Table S.3 the inner-clamped / outer-free set, both at $\nu = 1/3$. The $\nu = 1/3$ half of Table 2.22 contributes 40/40; its two WEAK cells are $(2,1)$ at $b/a = 0.5$ (1.006%) and $(3,0)$ at $b/a = 0.7$ (1.186%). The $\nu = 1/3$ half of Table 2.30 contributes 36/38 after the named $n = 0$ column at $b/a = 0.9$ is set aside; its two misses are the $(1,0)$ cells at $b/a = 0.1$ (10.806%) and $b/a = 0.3$ (3.635%), the same rows that miss at $\nu = 0.30$.

Table S.2: Mixed edges, inner free / outer clamped, $\nu = 1/3$, against SP-160 Table 2.22.

	$b/a = 0.1$		$b/a = 0.3$		$b/a = 0.5$		$b/a = 0.7$		$b/a = 0.9$	
(n, s)	printed	solver	printed	solver	printed	solver	printed	solver	printed	solver
$(0, 0)$	10.2	10.135	11.4	11.338	17.7	17.598	43.1	42.980	360	359.95
$(1, 0)$	21.1	21.189	19.5	19.392	22.0	21.785	45.3	45.078	362	361.04
$(2, 0)$	34.5	34.554	32.5	32.559	32.0	31.776	51.5	51.111	365	364.31
$(3, 0)$	51.0	50.992	49.1	49.108	45.8	45.487	61.3	60.573	370	369.75
$(0, 1)$	39.5	39.378	51.7	51.514	93.8	93.605	253	251.33	2219	2218.7
$(1, 1)$	60.0	59.999	59.8	59.362	97.3	97.040	254	253.36	2220	2220.1
$(2, 1)$	83.4	83.530	79.0	78.563	108.0	106.91	259	259.39	2225	2224.3
$(0, 2)$	90.4	90.161	132.0	132.14	253.0	251.94	692	691.70	6183	6184.4

S.4.3 In-plane free–free ring, all 48 cells

Table S.4 is the complete comparison with Irie, Yamada and Muramoto's Table 2: two frequencies per branch at each of four radius ratios, 48 cells, all MATCH.

Table S.3: Mixed edges, inner clamped / outer free, $\nu = 1/3$, against SP-160 Table 2.30. † named source-defect column (see main text). § search-window endpoint return; not a frequency and not printed.

	$b/a = 0.1$		$b/a = 0.3$		$b/a = 0.5$		$b/a = 0.7$		$b/a = 0.9$	
(n, s)	printed	solver	printed	solver	printed	solver	printed	solver	printed	solver
$(1, 0)$	3.14	3.4793	6.33	6.5601	13.3	13.313	37.5	37.566	345	345.47
$(0, 0)$	4.23	4.2629	6.66	6.7012	13.0	13.089	37.0	37.069	51.5†	§
$(2, 0)$	5.62	5.5139	7.96	7.8550	14.7	14.608	39.3	39.208	347	347.62
$(3, 0)$	12.4	12.230	13.27	13.050	18.5	18.310	42.6	42.383	352	351.22
$(0, 1)$	25.3	25.346	42.6	42.721	85.1	85.179	239	240.08	970†	§
$(1, 1)$	27.3	§	44.6	44.706	86.7	86.821	241	241.49	2189	2190.8
$(2, 1)$	37.0	36.925	50.9	50.941	91.7	91.766	246	245.71	2194	2194.5
$(3, 1)$	53.2	53.110	62.1	61.949	100	100.07	253	252.71	2200	2200.7

Table S.4: In-plane free–free ring against Irie, Yamada and Muramoto Table 2, $\nu = 0.3$, in the parameter of (S.5). Two frequencies per branch per β; all 48 cells MATCH, largest relative miss 0.222%.

	$\beta = 0.2$				$\beta = 0.4$			
	first		second		first		second	
branch	printed	solver	printed	solver	printed	solver	printed	solver
$n = 0$ radial	1.802	1.8015	4.748	4.7481	1.452	1.4521	5.529	5.5294
$n = 0$ torsional	3.090	3.0892	5.209	5.2085	3.530	3.5295	6.445	6.4452
$n = 1$	1.652	1.6512	3.842	3.8416	1.683	1.6824	4.044	4.0442
$n = 2$	1.110	1.1099	2.403	2.4024	0.721	0.7214	2.451	2.4501
$n = 3$	2.071	2.0707	3.401	3.3998	1.618	1.6185	3.346	3.3456
$n = 4$	2.767	2.7664	4.389	4.3875	2.482	2.4824	4.227	4.2264

	$\beta = 0.6$				$\beta = 0.8$			
	first		second		first		second	
branch	printed	solver	printed	solver	printed	solver	printed	solver
$n = 0$ radial	1.220	1.2197	7.979	7.9788	1.065	1.0647	15.754	15.7539
$n = 0$ torsional	4.867	4.8673	9.409	9.4087	9.380	9.3801	18.630	18.6299
$n = 1$	1.618	1.6178	5.092	5.0916	1.488	1.4879	9.458	9.4585
$n = 2$	0.418	0.4181	2.474	2.4737	0.178	0.1784	2.339	2.3389
$n = 3$	1.043	1.0430	3.433	3.4324	0.487	0.4870	3.299	3.2985
$n = 4$	1.752	1.7519	4.386	4.3850	0.892	0.8923	4.290	4.2895

S.5 Cluster job provenance

Every production number in the main text comes from a numbered batch job on The Mill. Main-text sections carry no job numbers; they are collected here. All jobs ran under `SOLVER_VERSION 2026-07-10.s10` with 40-digit working precision and a series truncation of $M = 80$ terms unless stated otherwise.

sweep	job
Free–free annulus, SP-160 Table 2.35 and Narita Table 2(a); fidelity anchors	`2461503`
Free disk, Narita Table 2(b), SP-160 Table 2.5, rank diagnosis, tiny-hole control	`2463651`
Finite elements, free–free annulus at five radius ratios and the free disk; mode identification from contours	`2461506, 2461507`
Mixed edges, SP-160 Tables 2.22, 2.30, 2.16 and Southwell Table 2.31; thin-ring diagnostic	`2463845`
Finite elements, mixed inner/outer edges at three geometries plus a refined mesh and a positive control	`2469112`
Secondary differential test against Vogel's table (§S.6.4)	`2469111`
In-plane free–free ring, Irie Table 2	`2461505`
Southwell inverse radius-ratio analysis and the window-endpoint audit	`container, no batch job`

The finite-element decks are ANSYS Mechanical APDL [2] with SHELL281 for flexure; the mixed-edge decks clamp one arc by radius selection and each wrote a non-zero clamped-node count, which is checked before the frequency listing is read. The queue script emits a cosmetic shell warning on every deck that is unrelated to the decks themselves.

S.6 Southwell's table: inverse radius ratio and predicted versus observed

This section supports the main text's claim that the MATCH/MISS split in SP-160 Table 2.31 is a sensitivity effect rather than an accuracy effect.

S.6.1 Method

Southwell's construction fixes round values of the Bessel argument and solves for the radius ratio that produces them, so in his table λ^2 is exact by construction and b/a is a computed quantity printed to two or three decimals. The forward comparison of the main text does the opposite: it holds the printed b/a fixed and asks what λ^2 the plate has. The inverse comparison holds the printed λ^2 fixed and root-finds on b/a.

For each of the eighteen rows the inverse search returns a CLEAN root with $s = 0$: 18/18. The resulting radius-ratio discrepancy $|\delta(b/a)|$ is then propagated forward through the local slope to a predicted relative error,

$$\text{pred} = \left(\frac{\mathrm{d}\lambda^2}{\mathrm{d}(b/a)}\right)\frac{|\delta(b/a)|}{\lambda^2}, \tag{S.6}$$

and compared with the observed forward miss. Slopes are evaluated numerically about each row's own operating point.

S.6.2 Result

Table S.5 is the full eighteen-row comparison. Two features carry the argument. First, *every* row — including the nine that MATCH — has a radius-ratio discrepancy of order 10^{-3}; over the thirteen MATCH and WEAK rows that quantity spans 1.6×10^{-4} to 6.9×10^{-3}, a band that overlaps the misses, so the discrepancy itself does not separate the two groups. Second, the ratio of predicted to observed relative error lies between 0.70 and 1.13 on all eighteen rows — within a factor of two, 18/18 — while the predicted error itself ranges over nearly three decades, from 7.3×10^{-4} to 0.45. A model that tracks a three-decade spread to within a factor of two is doing real work. The five direct misses are exactly the rows with the largest $\mathrm{d}\lambda^2/\mathrm{d}(b/a)$.

Table S.5: Southwell Table 2.31, inverse radius-ratio analysis. $|\delta(b/a)|$ is the difference between the printed radius ratio and the one that reproduces the printed λ^2 exactly; "pred" is (S.6); "obs" is the forward relative miss of the main text. ‡ the stored forward value at this row was a search-window endpoint; the observed error shown is recomputed from the wide re-search, which is CLEAN with $s = 0$. Ratio within a factor of two on all 18 rows.

n	b/a	printed λ^2	$\lvert\delta(b/a)\rvert$	$\mathrm{d}\lambda^2/\mathrm{d}(b/a)$	pred	obs	ratio	gate
0	0.276	6.25	0.00045	16.79	0.001206	0.00119	1.01	MATCH
0	0.642	25.0	0.00518	142.8	0.02959	0.02981	0.99	WEAK
0	0.84	81.0	0.04420	819.5	0.4472	$0.6419^{\ddagger}$	0.70	MISS
1	0.06	2.82	0.00016	30.89	0.001706	0.001838	0.93	MATCH
1	0.397	9.00	0.00066	31.72	0.002322	0.002299	1.01	MATCH
1	0.603	21.2	0.00054	109.8	0.002782	0.002746	1.01	MATCH
1	0.634	25.0	0.00049	140.3	0.002768	0.00274	1.01	MATCH
1	0.771	64.0	0.00134	570.4	0.01198	0.01191	1.01	WEAK
1	0.827	121.0	0.00490	1476	0.05979	0.05642	1.06	MISS
2	0.186	6.25	0.00435	10.53	0.007324	0.007153	1.02	MATCH
2	0.349	9.00	0.00119	24.98	0.00331	0.003267	1.01	MATCH
2	0.522	16.0	0.00018	64.48	0.000732	0.0007167	1.02	MATCH
2	0.769	64.0	0.00288	547.5	0.02465	0.02473	1.00	WEAK
2	0.81	100	0.00336	1079	0.03623	0.03474	1.04	MISS
3	0.43	16.0	0.00686	33.24	0.01425	0.01342	1.06	WEAK
3	0.59	25.0	0.00037	99.95	0.00149	0.001459	1.02	MATCH
3	0.71	49.0	0.01209	327.4	0.08081	$0.07457^{\ddagger}$	1.08	MISS
3	0.82	100	0.01008	1025	0.1033	$0.1104^{\ddagger}$	0.94	MISS

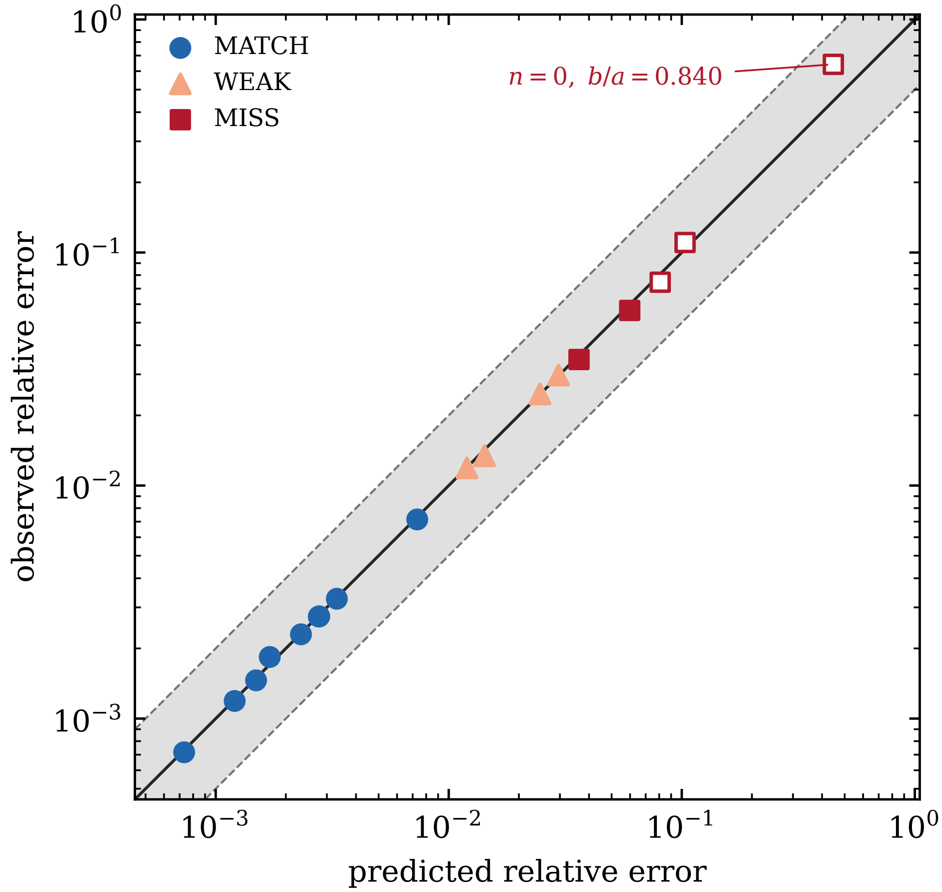

Figure S.1: Predicted versus observed relative miss on Southwell Table 2.31, all eighteen rows, log–log. The solid line is identity; the shaded band is a factor of two. Markers follow the same MATCH/WEAK/MISS gate as the main-text table. Open squares are the three rows whose stored forward value was a search-window endpoint; the observed error plotted is from the wide re-search (daggers in Table S.5). Ratio of predicted to observed error lies between 0.70 and 1.13 on 18/18.

S.6.3 Three rows were window endpoints

An audit of all 342 stored rows in this study against each row's own search window found twelve endpoint returns and no near-misses, all of them in the mixed-edge sweep. Three of them are Southwell rows, and all three had been counted as MISS, so correcting them does not change the 13/18 score. It does change the numbers, and the corrected values are what appear in the main text and in Table S.5:

cell	printed λ^2	stored (endpoint)	re-searched λ^2
$n = 0$, $b/a = 0.840$	81.0	—	**132.990**
$n = 3$, $b/a = 0.71$	49.0	—	**45.346**
$n = 3$, $b/a = 0.82$	100	—	**111.041**

The stored endpoint values are deliberately not reproduced: an endpoint is a property of the scanned window, not of the plate, and printing one beside a frequency invites it to be read as one. In the two $n = 3$ cases the true root lies just *outside* the half-window that was scanned, so the window was too narrow rather than mis-centred. Correcting the three rows improves the sensitivity result: the predicted-versus-observed ratio at $n = 0$, $b/a = 0.840$ falls from 3.13 to 0.70, which is what makes the unification 18/18 rather than 17/18.

The general lesson is recorded because it applies to any determinant search on a fixed absolute window: the sign-flip ladder caught all twelve of these (none classified CLEAN or FLOOR), but the MATCH/WEAK/MISS gate, being computed from the relative miss alone, did not. One counted cell — the $(1, 1)$ entry of Table 2.30 at $b/a = 0.1$ — was scored WEAK on an endpoint value, and its re-searched root is also WEAK (1.37% at $\nu = 0.30$, 1.56% at $\nu = 1/3$), so the reported pass rate is unchanged. It was, however, right by luck. Wiring the endpoint label into the scoring gate changes scoring and would need its own pre-registration; it has deliberately not been done for this paper.

S.6.4 A secondary differential test, and why it is not load-bearing

A separate test asked whether Southwell's residual radius-ratio discrepancies are larger than Vogel's when both are measured against the same model of what rounding alone should produce, namely half a unit in the last printed place of λ^2 divided by the local slope. The bar was fixed in advance at a ratio of three between the two medians, and the test passed: Vogel's median ratio of observed to expected discrepancy was 1.70, inside the calibration band, against 7.25 for Southwell.

It is nevertheless reported here as corroboration and not as evidence, for three reasons that were visible only after the run. The Vogel distribution is heavy-tailed — its upper quartile is 8.00 and 55 of 151 solved cells exceed a ratio of three — so the rounding model is an incomplete account of the residual for *both* tables, and comparing two medians drawn from distributions this dispersed is weak. There is an uncontrolled radius-ratio confound: Vogel's median ratio falls monotonically with b/a, from 9.70 at $b/a = 0.1$ to about 1.1 at $b/a = 0.9$, while Southwell's worst rows sit at large b/a and the two tables sample different ranges, so a purely b/a-dependent effect could reproduce the median difference with no source-precision story at all. And the inverse probe reproduced the same window-endpoint artifact in its own variable, returning one Vogel row at a radius-ratio offset exactly equal to its own half-window. The test was a real falsification opportunity that the sensitivity account could have failed and did not; the finite-element result of the main text is what settles the question.

References

[1] G. McCune, D. Stutts, "Exact wave-function solution and spurious-mode discrimination for completely free annular sector plates," companion paper, submitted (2026).

[2] ANSYS, Inc., *ANSYS Mechanical APDL*, Release 2025 R1, ANSYS, Inc., Canonsburg, PA, USA (2025).